

Exam 1 (Sept 21/25)

QTM 220

Instructions

Do Problem 1, either 2a or 2b, 3, and 4. No additional points will be awarded for doing both 2a and 2b; do whichever you prefer. If you do both, and it is not clear to us which one you want graded, we will grade Problem 2a.

Problem 1

Suppose we have a sample $Y_1 \dots Y_n$ stored on an **R** vector Y of length n . We run this code.

```
B=10000
Y.bar = rep(NA,B)
for(b in 1:B) {
  Y.bar[b]= mean(sample(Y,size=n,replace=TRUE))
}
```

Part a

What does this code calculate? State it in words and mark it with an ‘a’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

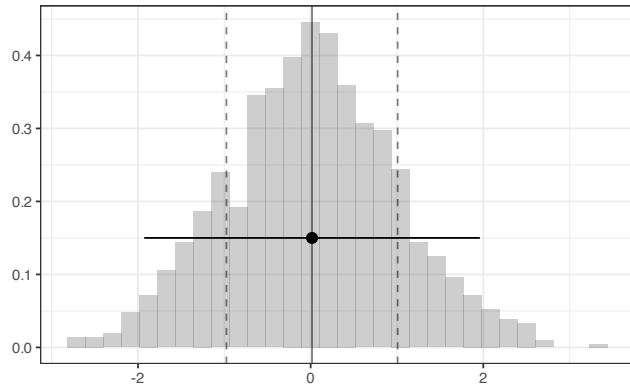


Figure 1: Histogram of the vector Y 's elements

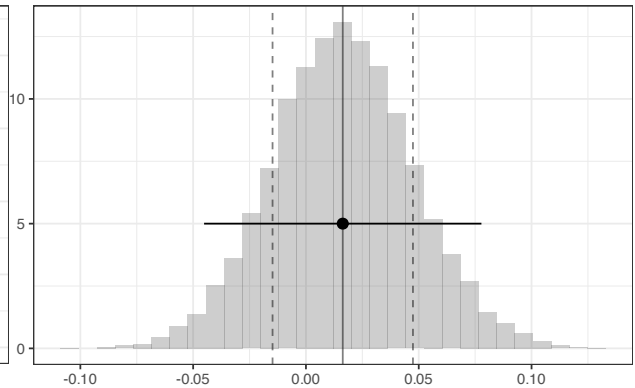


Figure 2: Histogram of the $\bar{Y}$'s elements

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

Part c

What does this code calculate? State it in words and mark it with an 'c', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y.bar)
```

Part d

What does this code calculate? State it in words and mark it with an 'd', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y) - 1.96* sd(Y)/sqrt(n)
mean(Y) + 1.96* sd(Y)/sqrt(n)
```

Part e

How could you calculate roughly the same thing as in part d using 'sd(Y.bar)' and not 'sd(Y)'?

Problem 2a

In 2015, Berkeley implemented a penny-per-ounce tax on soda. Advocates claimed the tax would reduce the rate of diabetes in the city's population. Now we want to check this claim—in a sense. What we want to know is the difference between the rate of diabetes among residents of Berkeley — one city in the San Francisco Bay Area — to the rate among residents of the Bay Area as a whole.

What we can use to do this is a sample, drawn with replacement, from the population of Bay Area Residents. To be precise, suppose the Bay Area has m residents, each with an id $j \in 1 \dots m$. Our population is $(x_1, y_1) \dots (x_m, y_m)$ where

$$x_j = \begin{cases} 1 & \text{if Bay Area resident } j \text{ lives in Berkeley} \\ 0 & \text{otherwise} \end{cases}$$
$$y_j = \begin{cases} 1 & \text{if Bay Area resident } j \text{ has diabetes} \\ 0 & \text{otherwise} \end{cases}$$

We have a sample $(X_1, Y_1) \dots (X_n, Y_n)$ drawn with replacement from this population.

Part a

Express your estimand — the population summary you want to estimate— in terms of $(x_1, y_1) \dots (x_m, y_m)$.

Part b

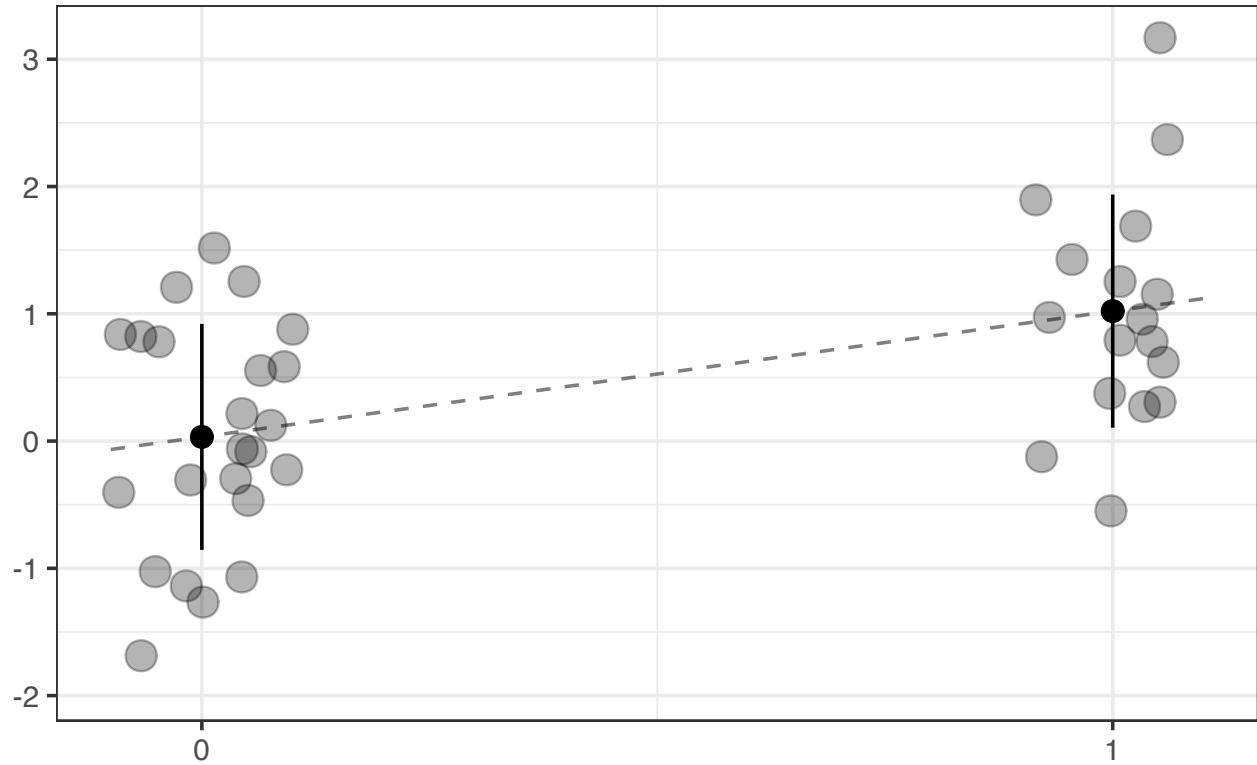
Express that estimand, using conditional and unconditional expectations, in terms of a single pair of random variables (X_i, Y_i) drawn from this population.

Part c

In terms of $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$, and $\overline{XY} = \frac{1}{n} \sum_{i=1}^n X_i Y_i$, state an *unbiased point estimate*.

Problem 2b

Suppose we have a bivariate sample $(X_1, Y_1) \dots (X_n, Y_n)$ stored in a pair of length- n vectors X and Y . And that $X_1 \dots X_n$ are binary, i.e., either 0 or 1.



Part a

What does the following code produce? State it in words and mark it with an 'a' on the plot above.

```
mean(Y[X==0])
```

```
[1] 0.03253742
```

Part b

Consider the least-squares regression line, $\hat{b}_0 + \hat{b}_1 x$ where

$$\hat{b}_0, \hat{b}_1 = \operatorname{argmin}_{a,b} \sum_{i=1}^n (b_0 + b_1 X_i - Y_i)^2.$$

We can get the coefficients $\hat{b}_0$ and $\hat{b}_1$ by running the following **R** code. **R** gives $\hat{b}_0$ the label ‘(Intercept)’ and $\hat{b}_1$ the label ‘X’.

```
regression_line = lm(Y~X)
coefficients(regression_line)
```

```
(Intercept)          X
0.03253742  0.98841557
```

Mark this line with a ‘b’ on the plot above. And explain why $\hat{b}_0$ is the same thing as your answer to the last question.

Part c

What does the following code produce? State it in words and mark it with a ‘c’, and an arrow or bracket if necessary for clarity, on the plot above.

```
mean(Y[X==1]) - mean(Y[X==0])
```

```
[1] 0.9884156
```

Part d

Let's revisit the least squares regression line introduced in part (b). Explain why the slope $\hat{b}_1$ is equal to the number we see in part (c).

Problem 3

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ drawn with replacement from a population with mean μ and standard deviation σ .

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n-2} \sum_{i=1}^n Y_i$.

Part a

What is $E[\tilde{Y}]$?

Part b

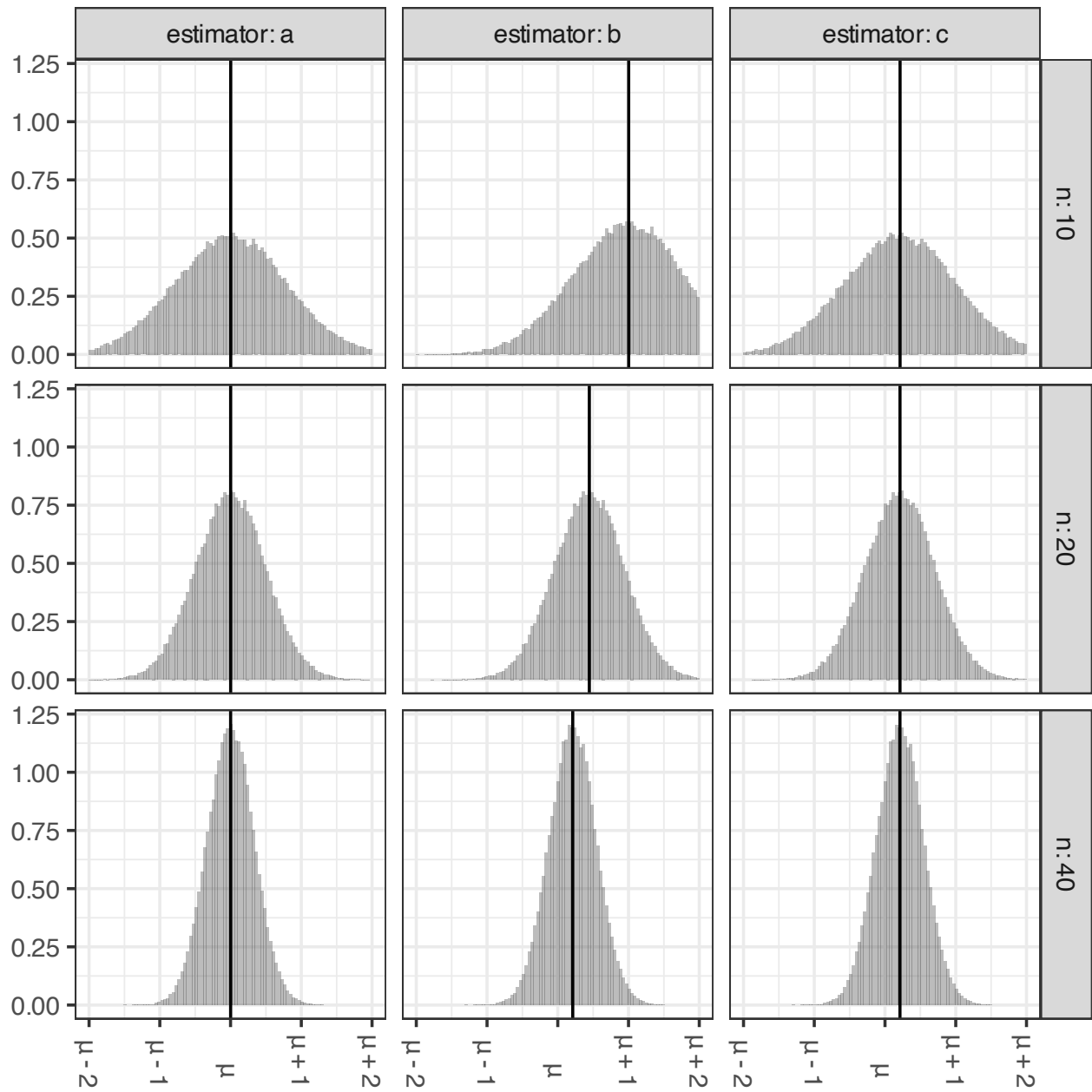
What is the bias of $E[\tilde{Y}]$ for μ ?

Part c

Showing your work, calculate $Var[\tilde{Y}]$.

Part d

The plot below shows the sampling distributions of three estimators at three sample sizes. The vertical line indicates the mean of the sampling distribution. Which of these estimators is $\tilde{Y}$? Is it a, b, or c?



Problem 4

Suppose a friend has polled 100 of Georgia's registered voters about their intention to vote in the 2024 election. The interval estimate they report for turnout is $[\cdot 59, \cdot 61]$. Could this interval estimate have 95% coverage? Explain your answer. *Hint. If Y is a binary random variable that takes on the value 1 with probability p (and 0 with probability $1 - p$), $V[Y] = p(1 - p)$.*